perception systems, drones are steadily evolving toward a level of situational awareness and adaptive behavior comparable to that of human pilots, enabling safe navigation in both structured and unpredictable environments.

7.3.3 Real-time decision-making with minimal human intervention

The hallmark of fifth-generation and emerging sixth-generation drones lies in their ability to perform real-time decision-making with minimal or no human intervention. This capability transforms drones from remotely controlled machines into autonomous, intelligent agents capable of analyzing data, assessing risks, and executing actions independently. Such autonomy is vital for missions requiring rapid adaptation, such as surveillance, search and rescue, logistics delivery, and combat operations, where communication delays or human limitations can compromise outcomes (Okere, et al. 2025).

Real-time decision-making depends on a combination of advanced onboard computing, artificial intelligence (AI), and sensor fusion technologies. Drones equipped with high-speed processors and AI-driven algorithms can analyze incoming data from cameras, LiDAR, radar, and inertial sensors to understand their environment and make immediate navigational or operational decisions. For instance, an autonomous drone can detect an obstacle, re-plan its route, and continue its mission all within milliseconds without operator input.

A key enabler of this autonomy is machine learning (ML), particularly through reinforcement learning and deep neural networks, which allow drones to learn optimal responses from past experiences and adapt to new scenarios dynamically. Edge AI computing, where decision algorithms run directly on the drone rather than relying on cloud processing, ensures